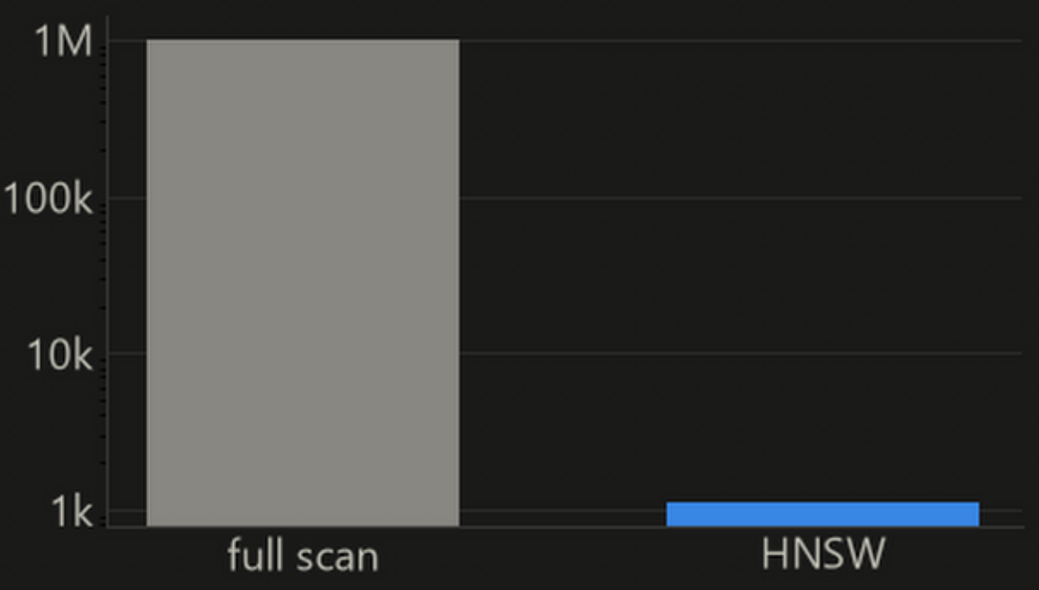
Nexus: a C++20 vector search engine, measured: 902x fewer distance computations than brute force, 500/500 crash recoveries verified

SIFT1M / SIFT-10K | single thread | 11,337 QPS at recall 0.951 on one core | worst-build recall after the graph fix 0.998 (was 0.64)

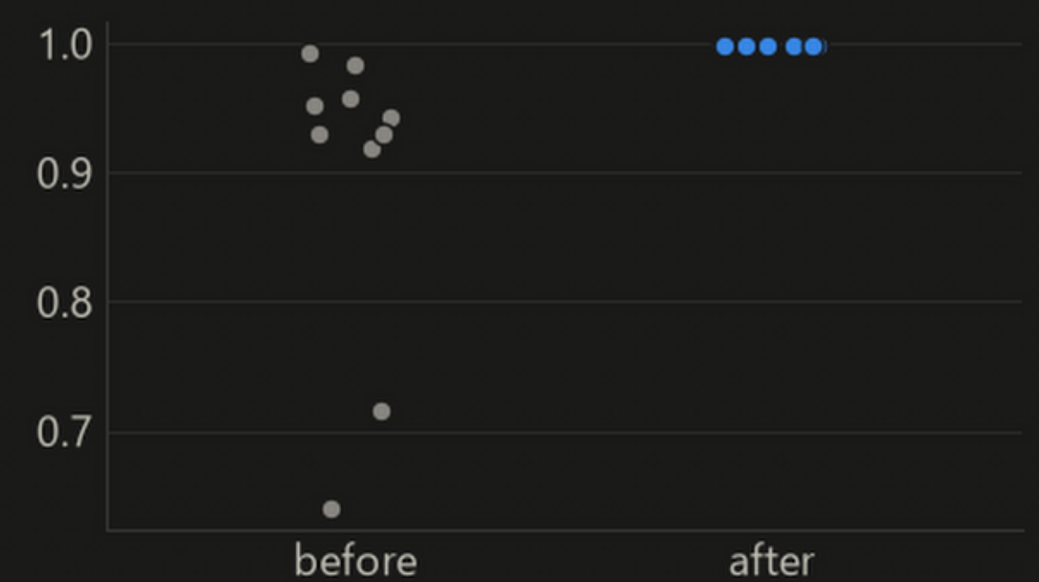
902x

fewer distance computations than a full scan at 95% recall (SIFT1M)



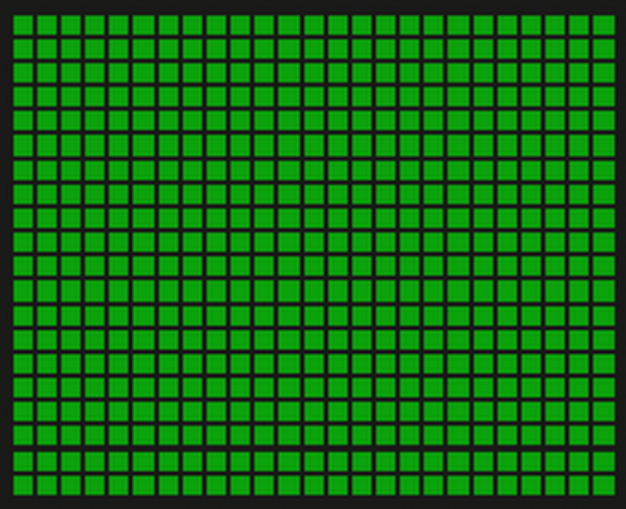
0.998

worst-build Recall@10 after the neighbor-selection fix (was 0.64)



500/500

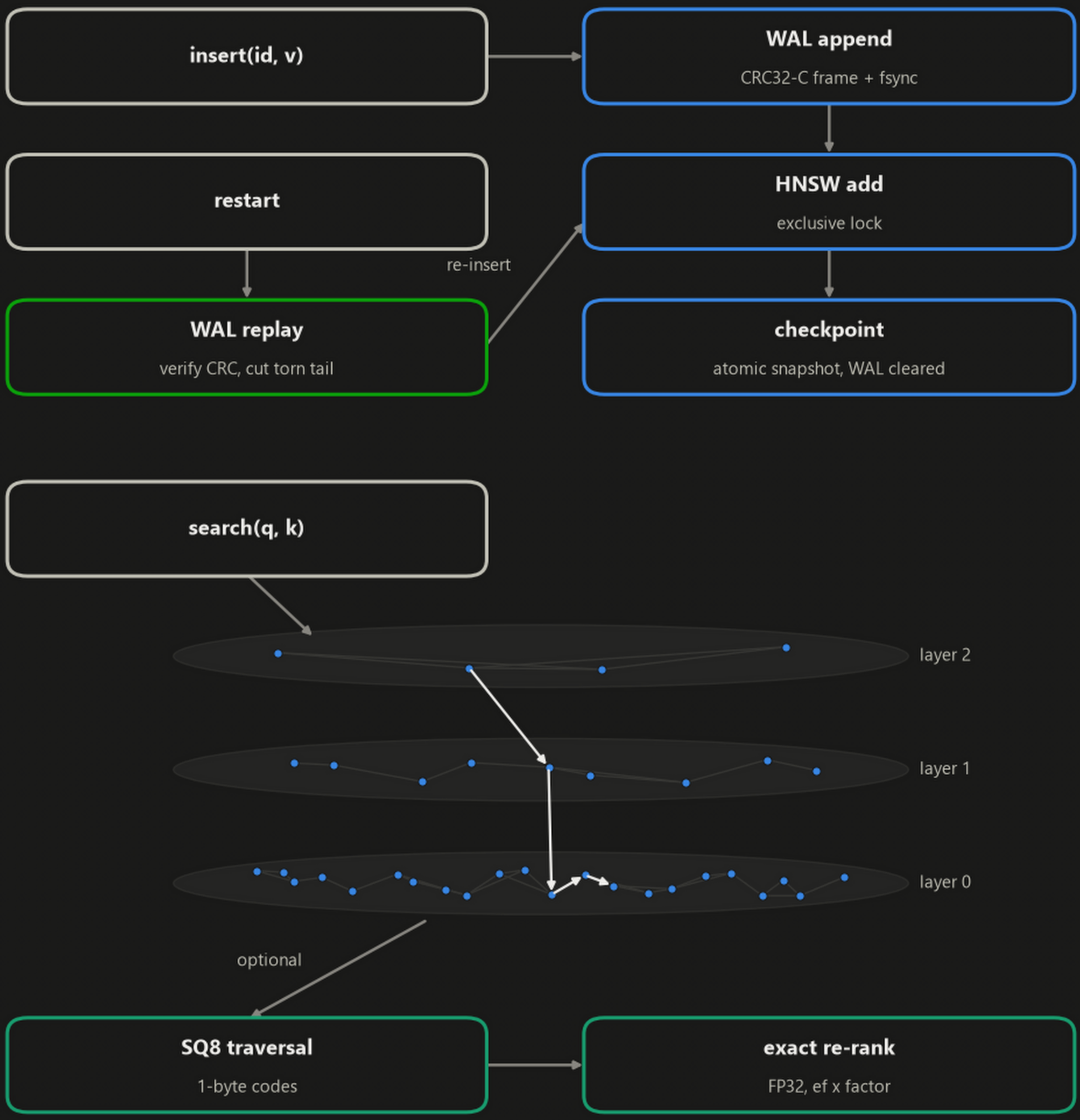
SIGKILL crash/restart cycles with 0 acknowledged inserts lost



intel-i5-13400f (13th Gen Intel(R) Core(TM) i5-13400F, wsl2, commit 6b47c62 + b48c33b) | all numbers measured, raw data in results/

How Nexus answers a query and survives a crash

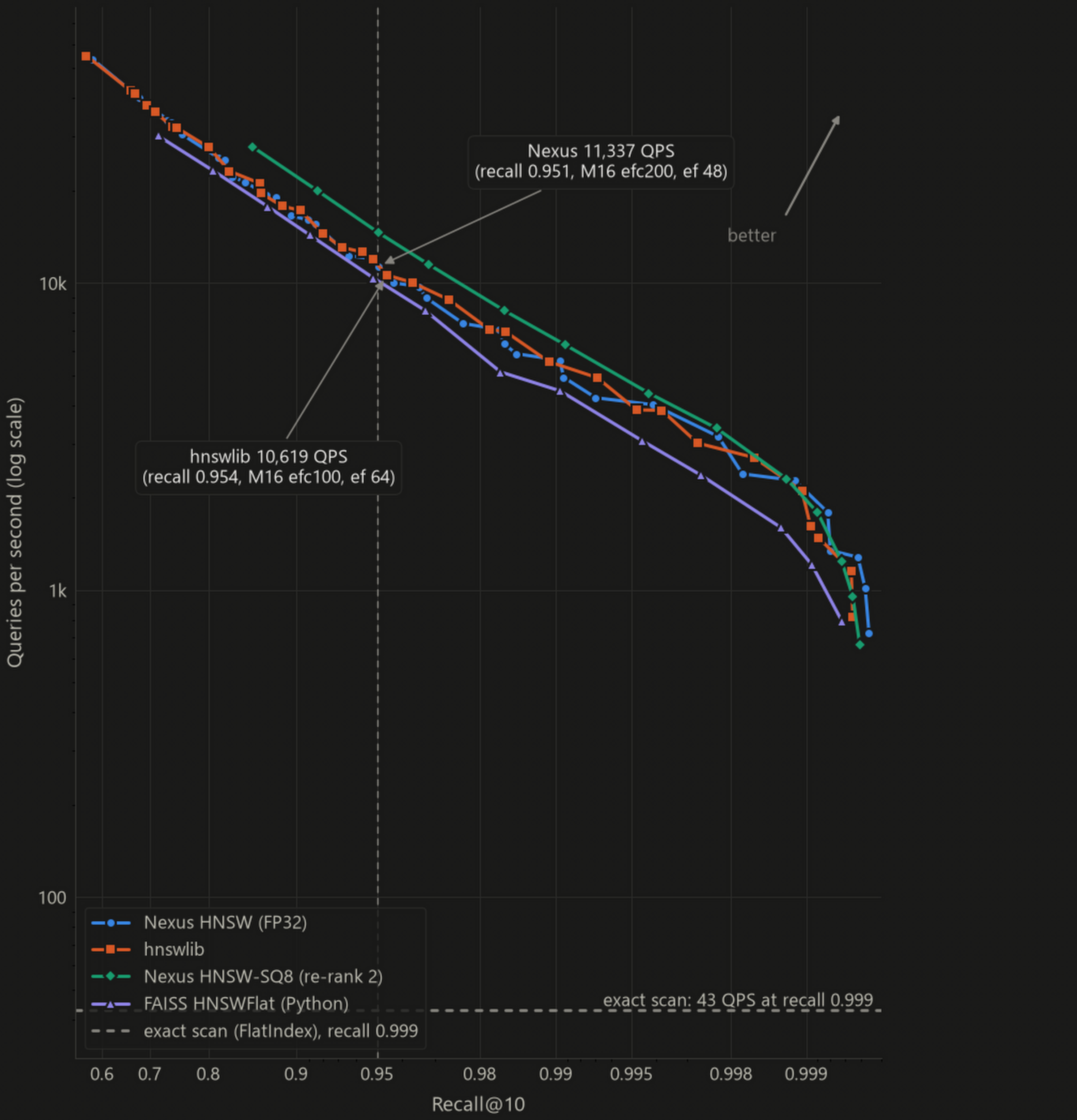
Writes: WAL record (CRC32-C, fsync) before the in-memory index; checkpoints snapshot the index atomically and truncate the WAL | Reads: greedy descent through the HNSW layers, beam search on layer 0, optional SQ8 traversal with exact FP32 re-ranking



intel-i5-13400f (13th Gen Intel(R) Core(TM) i5-13400F, wsl2, commit 6b47c62 + b48c33b) | all numbers measured, raw data in results/

At 95% recall on SIFT1M, Nexus answers 1.07x the queries per second of hnswlib

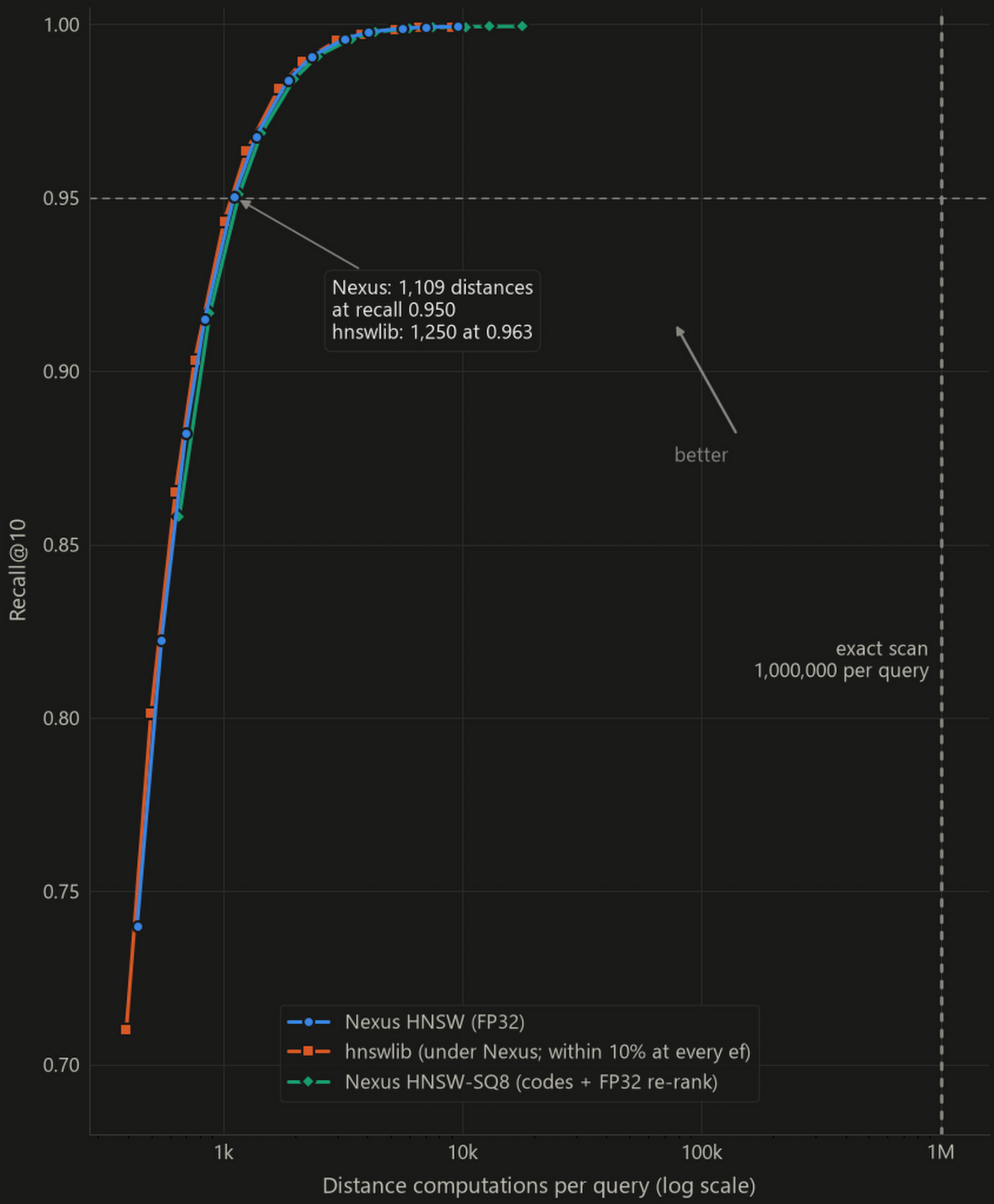
SIFT1M (1,000,000 x 128-d, L2) | k = 10 | best measured point of each engine across the same (M, efConstruction) grid | single thread, one query at a time



intel-i5-13400f (13th Gen Intel(R) Core(TM) i5-13400F, wsl2, commit 6b47c62 + b48c33b) | all numbers measured, raw data in results/

At 95% recall, HNSW computes 902x fewer distances per query than a full scan

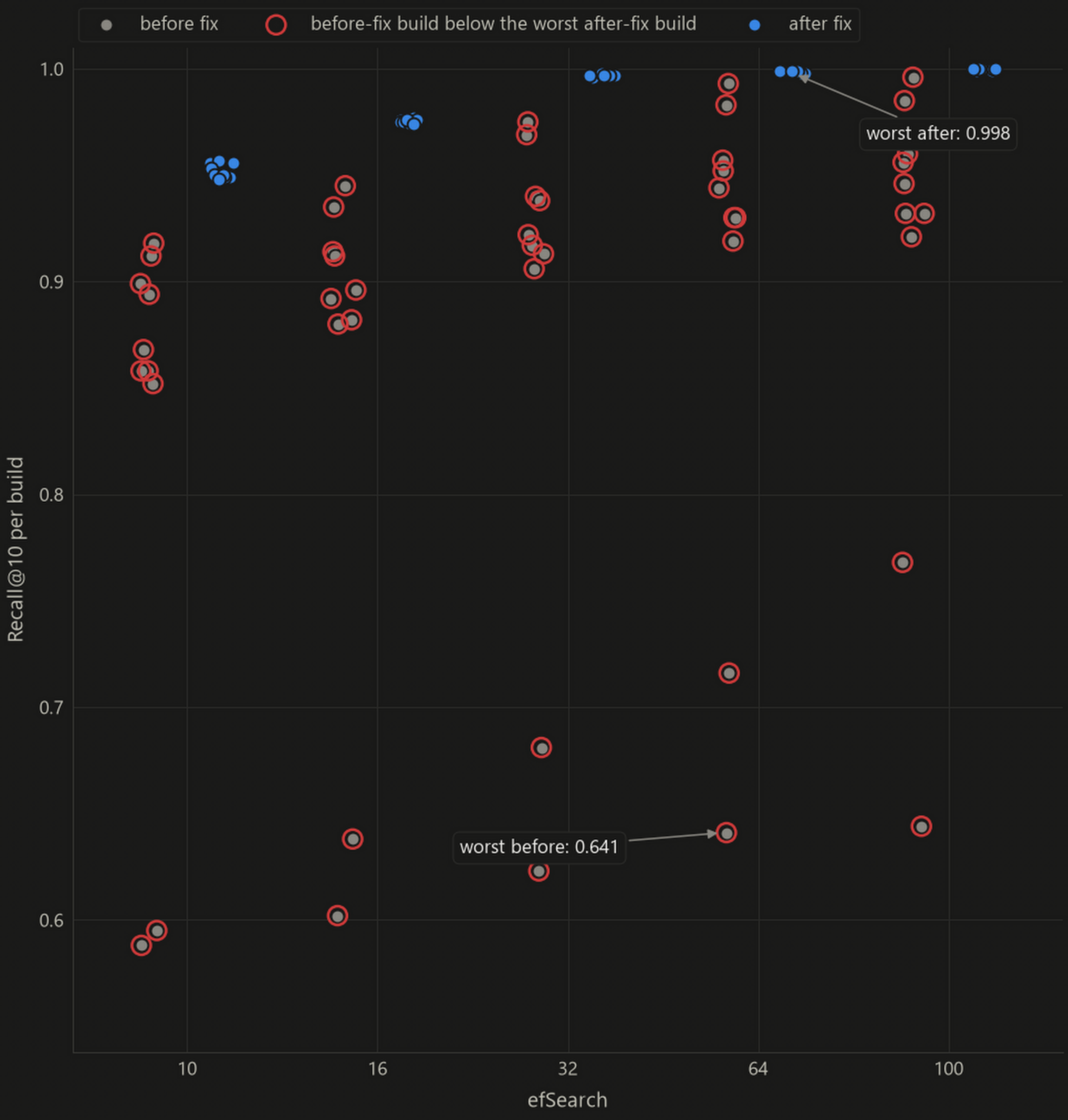
SIFT1M (1,000,000 x 128-d, L2) | k = 10 | counted with NEXUS_COUNT_DISTANCES (Nexus) and a counting wrapper around hnswlib's distance function, on the same saved graphs as the timing runs | hardware-independent measure



intel-i5-13400f (13th Gen Intel(R) Core(TM) i5-13400F, wsl2, commit 6b47c62 + b48c33b) | all numbers measured, raw data in results/

Neighbor-selection fix: worst of 10 builds went from recall 0.64 to 0.998 (efSearch 64)

SIFT-10K (10,000 x 128-d, L2) | Nexus HNSWIndex M=16, efConstruction=200, k=10 | 10 independent builds per version, each dot is one build | before = 414cadf (farthest-neighbor eviction), after = b48c33b (HNSW Algorithm 4 heuristic)

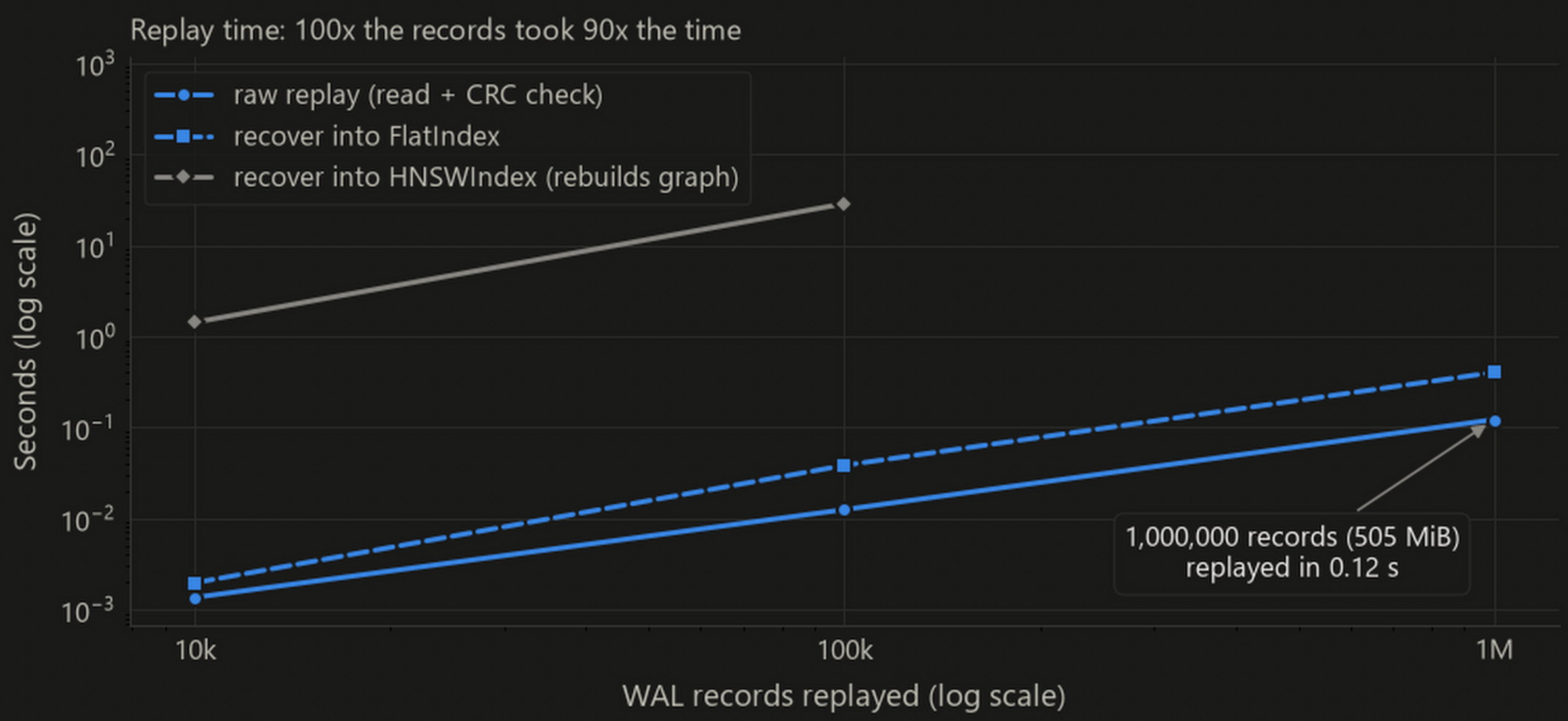


intel-i5-13400f (13th Gen Intel(R) Core(TM) i5-13400F, wsl2, commit 6b47c62 + b48c33b) | all numbers measured, raw data in results/

500 of 500 SIGKILL crashes recovered every acknowledged insert (0 lost, 125 torn tails truncated)

A writer inserts 128-d vectors (WAL append + CRC32-C + fsync per insert) and is killed at a random moment; after restart: no acknowledged insert missing, every vector bit-exact, torn tail truncated | every 4th cycle injects a partial record | WAL on ext4





intel-i5-13400f (13th Gen Intel(R) Core(TM) i5-13400F, wsl2, commit 6b47c62 + b48c33b) | all numbers measured, raw data in results/

HONEST LIMITS

What it does not do (yet)

- hnswlib is faster in 3 of 6 (dataset, recall-target) comparisons: SIFT1M >= 0.90: 1.08x; GloVe-100 >= 0.90: 1.05x; GloVe-100 >= 0.99: 1.05x.
- Graph construction on SIFT1M is 1.10x slower than hnswlib's, single-threaded (297 s vs 269 s); Nexus has no parallel build.
- HNSWIndexSQ8 uses more memory than FP32 HNSW (808 vs 680 bytes/vector), because it keeps FP32 copies for re-ranking.
- Durable writes are bounded by one fsync per insert (212 inserts/s here); there is no group commit.
- One machine (Intel i5-13400F) under WSL2; absolute numbers will differ elsewhere.

Code, raw data and how to reproduce:
github.com/orelshushan556-stack/Vector_Database_Engine